

Figure 6 — Matched means, by verdict

Matched dense-LIF forward, n = 20 per arm, 2026-08-25 re-run at MATCHED_INPUT_SCALE = 2.0. Every value is feed-forward / recurrent. Gate: primary mean ≥ 0.65 AND gap LCB > 0.5.

SuperSpike BPTT reference saturates at 1.0000 ff / 1.0000 rec. Against it every PASS below reduces to “the arm scored above 0.75”, so this figure does not rank them: rows are in the order of Table A.

Panel A — primary mean, grouped by verdict. Accuracy is printed, never encoded as a length or a position.

Does not clear the gate — the lead negative, and the only one

FAIL

broadcast ±1 three-factor

MatchedLocal — ±1 reward × surrogate eligibility. At chance on both graphs.

0.5000 ff / 0.5100 rec

Clears the gate — unordered, and not orderable against a saturated reference

PASS

graded DFA

MatchedDfaGradedError — graded error through fixed random feedback

0.9925 ff / 0.9875 rec

PASS

REINFORCE × frozen B_i

MatchedRIReinforceFb — the same ±1 reward, addressed

0.9950 ff / 0.9812 rec

PASS

discrete EventProp-style spike-adjoint

MatchedEventProp — the archived FAIL is WITHDRAWN; it had no spikes to differentiate through

0.9450 ff / 0.8900 rec

Measured and NOT gated — neither passes to be counted nor failures to be explained away

contrast (not gated)

broadcast graded error

MatchedBroadcastGradedError — why the lead negative says “±1 three-factor” and not “any broadcast”

0.9975 ff / 0.9975 rec

contrast (not gated)

RL graded-reward broadcast

MatchedRIGraded — richness without addressing

0.8787 ff / 0.9100 rec

contrast (not gated)

RL ±1 broadcast

MatchedRIFlat — same reward and topology as the lead FAIL, 0.28 above it

0.7775 ff / 0.7962 rec

The ceiling every arm above is measured against

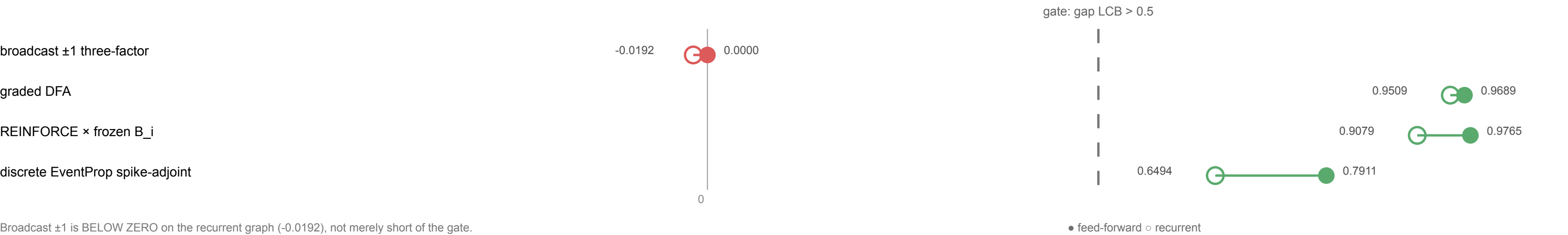
reference

SuperSpike BPTT

saturates in every suite, on both forward graphs

1.0000 ff / 1.0000 rec

Panel B — gap LCB against the > 0.5 gate. Same row order; the second half of the gate, drawn because clearing the floor and clearing the gate are different sentences.



Broadcast ±1 is BELOW ZERO on the recurrent graph (-0.0192), not merely short of the gate.

Disclose with this figure: the broadcast-GRADED contrast reaches 0.9975 and ±1 broadcast REINFORCE reaches 0.7775. Neither is a PASS and neither erases the ±1 × eligibility FAIL.